

CareSense AI

Smart Healthcare Assistant for Elderly Monitoring

Problem Statement Document

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1. Domain

Healthcare – AI-Powered Elderly Health Monitoring & Preventive Care

This solution sits at the intersection of digital health, ambient assisted living, and conversational AI. It targets the growing global need for scalable, low-friction elderly care systems that reduce dependency on constant human supervision while keeping caregivers informed and in control.

2. The Challenge

Elderly individuals — particularly those living independently, in assisted-living facilities, or with limited caregiver availability — require continuous monitoring across several dimensions of physical wellbeing:

- **Posture and prolonged sitting/inactivity**, which increases risk of pressure sores, poor circulation, and musculoskeletal issues.
- **Vital signs** such as heart rate, SpO2 (blood oxygen saturation), and body temperature, which can signal early onset of cardiac, respiratory, or infectious conditions.
- **Fall risk**, one of the leading causes of injury-related hospitalization and mortality in the elderly population.

Today’s solutions largely fall short in two ways:

1. **Fragmentation** – posture tracking, vital monitoring, and fall detection typically live in separate devices or apps with no unified view, forcing caregivers to piece together a picture of the patient’s condition manually.
2. **Lack of real-time preventive guidance** – most systems are reactive (alerting only after an adverse event) rather than proactive (flagging risk patterns before they escalate), which delays timely caregiver intervention and reduces the window for preventive action.

This gap creates both a safety risk for elderly users and an operational burden for caregivers and healthcare providers who must monitor multiple data streams without a consolidated, intelligent assistant.

3. The Objective

Build an **AI-powered Smart Healthcare Assistant** using **IBM watsonx Orchestrate** that unifies health data monitoring, delivers personalized preventive recommendations, and keeps caregivers informed — all through a coordinated multi-agent architecture rather than a single monolithic model.

The system should be able to:

- Continuously ingest and interpret multi-modal health signals (posture, vitals, activity duration).
- Contextualize readings against the individual’s baseline and known risk factors.
- Proactively recommend preventive actions before a condition becomes acute.
- Escalate to caregivers with clear, actionable alerts rather than raw data dumps.
- Present all of this through an accessible, easy-to-read dashboard suitable for both elderly users and their caregivers.

4. Key Features

Feature	Description
Personalized Health Analysis	Builds an individual baseline for each user (normal HR/SpO2/temperature ranges, typical activity patterns) and flags deviations specific to that person rather than generic thresholds.
Health Knowledge Retrieval (RAG-ready)	Uses Retrieval-Augmented Generation to ground recommendations in verified medical knowledge bases, ensuring advice is accurate, current, and explainable rather than hallucinated.
Health Monitoring & Feedback	Continuously processes incoming sensor/vitals data and returns real-time, human-readable feedback (e.g., “You’ve been seated for 90 minutes — consider a short walk”).
Preventive Healthcare Recommendations	Suggests posture corrections, hydration reminders, activity breaks, and lifestyle nudges tailored to the user’s risk profile, aiming to intervene before issues arise.
Caregiver Alerts & Health Reports	Generates prioritized alerts for caregivers when readings fall outside safe ranges or fall risk is detected, along with periodic summarized health reports.
Interactive Health Dashboard	A web-based (HTML/CSS/JS) interface visualizing real-time vitals, trends over time, and agent recommendations for both the elderly user and their caregiver.

5. Multi-Agent System

Rather than a single AI handling every task, CareSense AI is designed as a coordinated team of specialized agents orchestrated through IBM watsonx Orchestrate:

5.1 Health Data Analysis Agent

Responsible for ingesting raw sensor and vitals data, cleaning/normalizing it, detecting anomalies against personal baselines, and passing structured insights downstream to the other agents.

5.2 Health Recommendation Agent

Consumes the analyzed data and generates specific, actionable preventive recommendations (posture correction, movement breaks, hydration, sleep hygiene) tailored to the user’s current state and history.

5.3 Health Advisory Agent

Handles knowledge-grounded Q&A and general health guidance using RAG over a curated medical knowledge base, allowing the elderly user (or caregiver) to ask natural-language health questions and receive reliable, sourced answers.

5.4 Caregiver Support Agent

Monitors for threshold breaches and risk escalations, composes and dispatches caregiver alerts, and compiles periodic health reports summarizing trends, incidents, and adherence to recommendations.

These agents communicate through the Orchestrate framework, allowing each to specialize while contributing to a single coherent user experience.

6. Technology Stack

Layer	Technology
Agent Orchestration	IBM watsonx Orchestrate
Foundation Models	IBM watsonx.ai, IBM Granite Models
Knowledge Retrieval	RAG (Retrieval-Augmented Generation)
Vector Storage (<i>future</i>)	Vector Database
Frontend	HTML, CSS, JavaScript
Infrastructure	IBM Cloud

7. Future Scope

The initial software-only build is designed to evolve into a full hardware-integrated care ecosystem:

- **Smart Elderly Care Chair** — an ESP32-based physical device instrumented with IoT sensors (posture, pressure, motion) to feed real-time physical data directly into the agent pipeline.
- **Firestore Integration** — real-time database sync between the chair’s sensors and the cloud-based agent system for low-latency updates.
- **Predictive Analytics** — moving from reactive/preventive alerts to predictive risk modeling using historical trend data (e.g., forecasting fall risk before it manifests).
- **Caregiver Mobile Application** — a dedicated companion app for caregivers to receive push alerts, view dashboards on the go, and manage multiple patients/family members from a single interface.

8. Expected Outcomes

- A working multi-agent prototype demonstrating end-to-end flow: sensor/vitals input → analysis → recommendation → caregiver alerting → dashboard visualization.
- Demonstrated reduction in “time-to-alert” for risk conditions compared to manual/fragmented monitoring.
- A foundation that can be extended with real IoT hardware (Smart Care Chair) without re-architecting the core agent logic.